International Journal of Disaster Risk Reduction 51 (2020) 101881

Contents lists available at ScienceDirect

International Journal of Disaster Risk Reduction

journal homepage: http://www.elsevier.com/locate/ijdrr

Review Article

Knowledge management practices in disaster management: Systematic review

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ARTICLE INFO

Keywords:
KM practices
Natural hazards
Disaster
Implication

ABSTRACT

Examples of knowledge management (KM) practices are common in a disaster context. However, disaster management is a system with many components, and it is not clear how KM practices can be applied in all major phases of disaster management cycle. In this context, this research will answer the following questions: i) What are some of the KM practices used in disaster management? and ii) What are some of the implications of KM practices on disaster management performance? Moreover, the key findings of this study acknowledge the recognition of research gaps that need to be addressed and they can provide future researchers with a research agenda. This article provides a systematic review of the 72 selected papers from SCOPUS databases on KM in the context of disaster management to evaluate the state of the art, finding existing gaps and establishing a direction for further studies. This review explores the use of KM in all phases of disaster management to confirm the advantages of knowledge management in reducing disaster impacts and enhancing disaster resilience.

1. Introduction

The number of disasters that are increasingly reported each year during the past decade become more severe and they affect more people. In 2018, a higher proportion of disasters were reported worldwide. From a total of 315 disaster events caused by natural hazards, the number of deaths reached 11,804, with 68 million people being affected and 131,7 billion USD of economic damage. Indonesia reported almost half the total deaths (47%), while India ranked first in number of people affected (35%) [1]. As the number and variety of disasters that can cause direct and indirect effects have increased in the world, many countries are increasingly concerned with the pressing need to reduce the disaster risk and to develop a resilient disaster community.

Each year, millions of people suffer seriously from disasters as the result of insufficient coping mechanisms, or such mechanisms exist but they are not transformed into life-saving knowledge regarding endangered communities. This is one of the major causes underpinning the current poor achievement of disaster management practice. In particular, this indicated a lack of knowledge sharing and creation of effective disaster management strategies [2].

Knowledge management means gaining the correct knowledge at the appropriate time and place. It aims to facilitate the process of creating, sharing and making use of knowledge. Although disaster-related negative consequences cannot be neutralized, efforts can be affected in order to minimize their impacts, and knowledge of disaster management strategies, good practices and learned lessons, will undoubtedly support this effort by means of informed mitigation actions and preparedness planning.

In order to implement best practices during the whole disaster management cycle, disaster management practitioners must be

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https://doi.org/10.1016/j.ijdrr.2020.101881
Received 30 January 2020; Received in revised form 25 August 2020; Accepted 16 September 2020
Available online 22 September 2020
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innovative and they must learn from the lessons that are taught in practical and real-time situations. In brief, practitioners must constantly enhance their knowledge and skills. To achieve this end, investments in systems, databases and network structures need to be made to develop a culture of learning from prior knowledge and also from current best practices [3].

However, to our knowledge, there is fragmented information regarding disaster management strategies, thus stressing a perceived lack regarding the coordination and sharing of information. Consequently, disaster practitioners' knowledge and experiences remain at the individual or institutional levels. Therefore, the experience, approaches and methods adopted for managing disasters remain a tacit knowledge for individuals. This led to the reinvention of the wheel in the establishment and management of programs and projects before and after a disaster.

Examples of KM practices are common in the disaster context. However, disaster management is a system with many components, and it is not clear how KM practices can be applied in all major phases of disaster management cycle. Moreover, to our knowledge, there has no systematic analysis of the literature on the application of KM in disaster management.

This review explores the use of knowledge management in all phases of disaster management phases to confirm the advantages of knowledge management in reducing disaster impacts and enhancing disaster resilience. In this context, this research will answer the following questions: i) What are some KM practices used in disaster management? and ii) What are some implications of KM practices on disaster management performance? Moreover, the key findings of this study acknowledge the recognition of research gaps that need to be addressed and they can provide future researchers with a research agenda.

This paper is developed in six sections. The principles of KM and disaster management are discussed in section 2. The review protocol that has been carried out is discussed in section 3. General statistical information and analysis from the literature review are provided in section 4. The future research agenda based on key findings is discussed in section 5. The conclusions of this research are discussed in section 6.

2. Concept of Knowledge Management and Disaster Management

2.1. Knowledge management

Knowledge management facilitates the collective and systematic creation, distribution and utilization of knowledge by individuals, teams and entire organizations to reach their strategic and operational goals. Knowledge management leads, on the one hand, to improving the effectiveness and performance of operations and, on the other hand, to innovating and improving the level of competition. Knowledge-based management aims at creating information knowledge and turning this expertise into a sustainable competitive advantage so that it becomes part of business success [4].

As a scientific discipline, KM has been developing for more than 30 years, which is not only studied in academic theory but also becomes an important component in organizational life. The term KM has been applied to various fields of science to manage, create, and increase intellectual assets [5]. Besides developing in the management studies field, KM is also discussed within other disciplines, some of which are dominant include information technology, communication, philosophy, cognitive science, social science, artificial intelligence, and economics [6].

Nonaka and Takeuchi (1995) define KM as a management tool that makes individual knowledge an asset with which to enhance organizational effectiveness through dynamic interactions [7]. KM is a management function that is aimed at strategies of formulation, implementation, and evaluation which can guarantee the process of disseminating the appropriate knowledge and format, to the right individuals, at the correct place and time [8]. Serrat [9] defines KM as a combination of explicit and systematic information management with learning organizations that allow the identification, creation, storage, sharing and utilization of knowledge resources, both individual and collective.

The implementation of knowledge management can be divided into knowledge management based on Information Technology (IT), people and processes. Knowledge management on the basis of IT and/or supply highlights the need for (facility) access to current knowledge in stored in databases or other places. From a human-centred viewpoint, the focus of knowledge management on (documented) knowledge collections has been expanded to include people-to-people connections that accommodate social relations with their corresponding technology support (i.e., environments of customary knowledge) [10].

Based on the above, there are at least four elements used by some researchers in defining KM practices, namely: i) People-oriented KM, ii) Process-oriented KM, iii) Technological-oriented KM, and iv) Goal-oriented KM [7,8,11].

The previous studies suggested that people-oriented KM practices are highly associated with innovation, and several studies note that the use of people-oriented KM improves the knowledge processes (i.e., acquisition, distribution, and creation), which influence on innovation skill. Notwithstanding, people-oriented KM practices appeared to stimulate and promote a sense of innovative by owning a positive impact on the individual's affective engagement and impersonal trust. Strategic are primarily associated with innovations. If all of the necessary management process-oriented KM practices were actually in place, it would advance the organization's performance. Some of the examples of process-oriented KM practices include the concept of KM for management at the top of the pyramid, a broadness of knowledge strategy targets and KM tools, and implementation support elements (i.e., cultural principles and leadership) [4,10].

Technology-oriented KM support for individuals who collaborate, communicate, undertake information searches, participate in real-time learning, together with simulating predicting is associated with the innovativeness of a company. Researchers also noted that technological-oriented KM is a primary driving of knowledge acquisition, knowledge creation, and knowledge sharing, which accelerates organizations towards improving their performance via innovations and organizational readiness. Goal-oriented KM has an orientation to promote the performance of both individuals and organizations by ensuring the appropriate knowledge, can be used in the right format and manner as well as at the right time and place [4,10].

2.2. Disaster management

Disaster is defined as a severe disruption to the work of an area or society at any level because of dangerous events interacting with exposure conditions, vulnerability, and ability that lead to the following: loss and impacts in human, material, economic and environmental areas [12]. The disaster also refers to a sudden, disastrous event that severely has disrupted the workings of a community or a society and causes losses that are greater than the capacity of the community or society to cope with using their own resources. It also leads to material and economic losses [13]. It can be determined from both definitions, that disasters can occur due to i) the interaction between hazard, vulnerability, and capacity; ii) serious damage in terms of human, material, economic and environmental aspects; and iii) causing disruption to people's lives that exceeds the local capacity to manage with using their own resources and hence there is a call for external support.

Disaster management is considered as a key factor and an effective strategy to minimize the impact of any disaster. Disaster management is a strategic planning and a procedure that is administered and employed to protect vital infrastructures (i.e., critical assets) from severe damages when man-made calamities and natural catastrophic disasters occur. As aforementioned, as a systematic process Disaster Management leads to

Table 1
Research keywords.

Group	Search Terms	Justification
Group 1	"knowledge management", "knowledge acquisi*", "knowledge creat*", "knowledge conver*", "knowledge storage*", "knowledge sharing", "knowledge applica*", "knowledge assets*", "community of practice*"	To identify literature related to knowledge management element
Group 2	"disaster management", "prevent*", "mitigat*", "preparedness*", "emergency response*", "relief*", "rehabilitat*", "reconstruct*", "recovery*"	To identify literature related to disaster management phases
Group 3	"earthquake*", "tsunami*", "flood*", "landslide*","drought*", "hurricane*", "volcan*", "storm*"	To identify literature related to type of natural hazards

action pre-disaster, during-disaster and post-disaster in order to save human lives [12].

Before a disaster, it involves mitigation and preparedness activities. Mitigation is to a series of efforts to minimize any risk of disaster via physical development, awareness, and capacity building to face the threat of disaster. Preparedness is a series of activities that are affected in order to anticipate disasters through organizing and appropriate and effective steps. During a disaster, it covers an emergency response phase. Emergency response is a series of activities that are effected at the moment a disaster occurs in order to manage any adverse impacts that are created, which include 1) victim rescue/evacuation, 2) safety of property, 3) fulfilment of basic needs, 4) rescue, protection and management of any refugees, and 5) restoration of infrastructure and facilities [12].

Post-disaster involves two stages: 1) Rehabilitation: improvement/restoration of the entirety of public/community services to a reasonable level for the post-disaster areas with the primary objective of the normalization or regular operation of all aspects of government and community life in post-disaster areas; 2) Reconstruction: the rebuilding of all post-disaster area infrastructures/facilities/institutions at government/community levels with the main aim being the growth and development of all activities (i.e., economic, social and cultural), resurrecting law and order, and the augmentation of community participation in all areas of community life in the region, post-disaster [12].

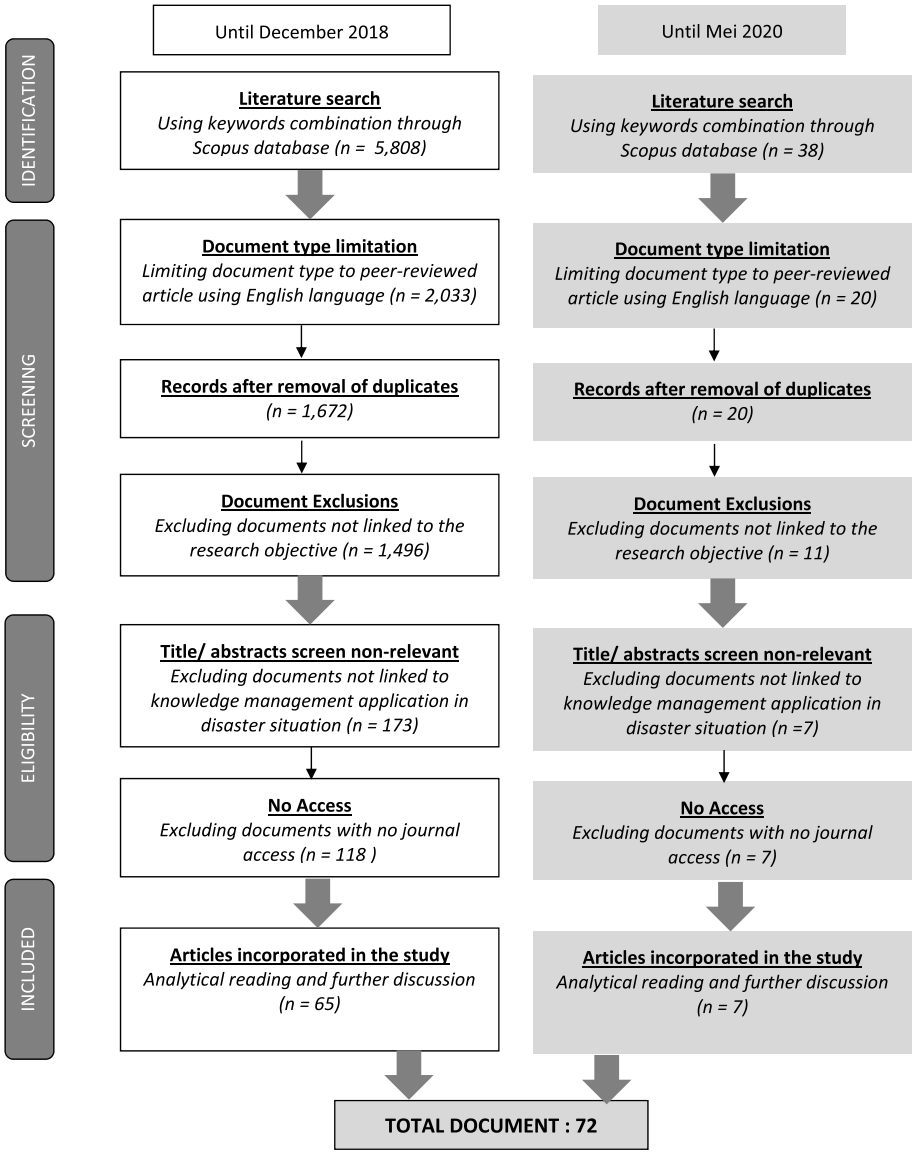


Fig. 1. PRISMA flow diagram of systematic review.

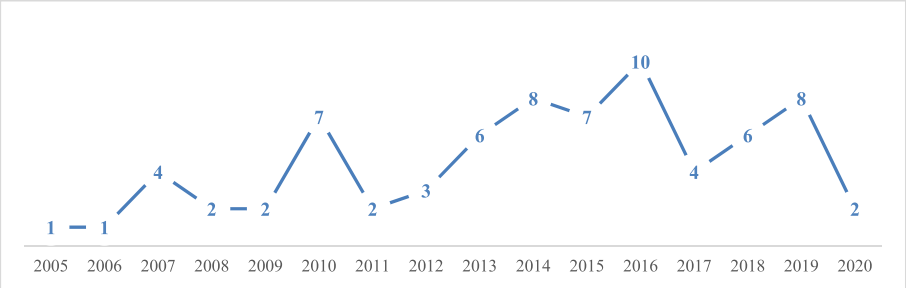


Fig. 2. Growth of publications by year.

3. Materials and methods

This study employed a systematic literature review (SLR) to present a comprehensive, exhaustive summary of current evidence relevant to research questions. An SLR is an overview of scientific contributions on a specific subject affirming replicable methods. SLR has a ten-stage organization, starting with identifying keywords to the point validating any selected papers via the citation technique This study employed a systematic literature review (SLR) to present a comprehensive, exhaustive summary of current evidence relevant to research questions. An SLR is an overview of scientific contributions on a specific subject affirming replicable methods. SLR has a ten-stage organization, starting with identifying keywords to the point validating any selected papers via the citation technique [14]. Petticrew and Roberts [15] introduce a conceptualization of SLR as a "review that attempts to comprehensively identify, assess and synthesize all associated studies on an assigned subject" and proposed a review arranged into 12 steps. Easterby-Smith et al. [16] classify two main processes in carrying an SLR, particularly: i) Define the review protocol and the relevance of the research in the field that is being studied, and ii) Explication of the main study findings with the aim of identifying knowledge and research gaps in the existing body of literature.

Scopus database was chosen to perform this study because it has a more significant number of indexed journals [17,18]. The research keywords were applied to titles, abstracts, and keywords, being sufficiently broad-based so that they would avoid any artificial limitation on the papers/books that were retrieved, but nevertheless maintaining limits that would exclude undesirable results. The research keywords are reported in Table 1.

The literature was primarily obtained by combining the keywords in "group 1 and group 2" and "group 1 and group 3" with the Boolean operators "AND" and "OR", asterisk (*) and the quotation marks ("). The search was conducted between December 2018 and Mei 2020.

This review intends to evaluate the research focused on KM practices in disaster management. Thus, only peer-reviewed articles published in a journal using the English language were taken into account. Documents that did not meet the purpose of the review and were not directly related to knowledge management in disaster situations were excluded from this study. Besides, papers with no full article were not included in further analysis.

4. Results and discussion

The search procedure identified 5808 articles, of which 1672 were left outstanding after we had removed any duplicates. After applying the inclusion/exclusion criteria from titles and abstracts scanning, 174 remained; for 119 cases complete manuscripts were accessible.

From the assessment of the full manuscript, 66 articles have been selected for further discussion and analysis. Another six selected articles published until Mei 2020 were identified to arrive at 73 included articles for review. The flow diagram for the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA), in accordance with Moher et al. [19] are presented in Fig. 1.

Table 2
Number of articles according to the type of natural hazards.

Type of natural hazard	No of documents
Earthquake	7
Tsunami	2
Earthquake and Tsunami	2
Flood	20
Tsunami and Flood	2
Flood and Landslide	3
Drought	2
Hurricane/cyclone/typhoon	4
Flood and hurricane/cyclone/typhoon	1
Volcano	1
Earthquake, flood, drought and hurricane/cyclone/typhoon	1
Not specified	27
TOTAL	**72**

4.1. General statistic information

According to the number of publications over time (Fig. 2), a notable number of articles were issued in 2016 (10 publications). In addition, there was only one paper in 2005 and 2006. As a result, the publication trend on the subject of KM practices in the disaster field has been increasing in recent years.

Table 2 highlights the number of publications categorized by the type of natural hazards. There were twenty-five articles that were drawn from the experience of a flood disaster. This number could be related to the frequency of flood events that occur each year. Throughout the start of the 21st century, more people were affected by floods than any other type of disaster [1]. Another 27 articles did not explicitly write down the types of natural hazard events.

The analysis of keywords co-occurrences was conducted using the VOS viewer software. Circles drawn on the map represent a number of co-occurrences of keywords. The bigger the circle, the higher the keyword's occurrence number. Keywords are clustered in groups marked with one colour based on their year of publication, with colours ranging from dark blue (published before 2010) to yellow (published after 2016).

However, the closer the circles are to each other, the more often the words have fallen together, and the more extensive the line connecting them – the more significant the co-occurrence. Accordingly, Fig. 3 shows that keywords such as disaster management, decision making, disaster prevention, and disaster risk reduction occurred in the study most frequently.

4.2. Content analysis of the literature

The results of the literature analysis show that, in general, KM practices have been performed in disaster management. However, the majority of examples in the literature only provide fragmented insights

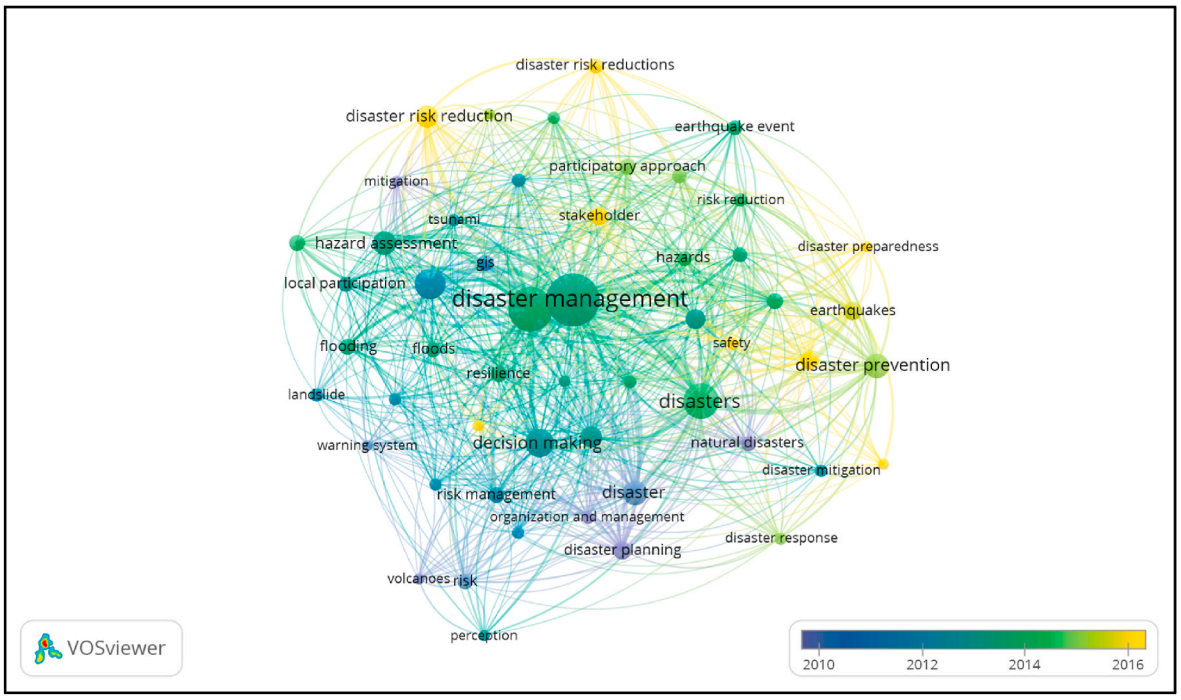


Fig. 3. Mapping of co-occurrences of keywords.

of KM practices in managing a disaster. To the best of our knowledge, there is no single article that comprehensively executes KM practices using people-, process-, technology- and goal-oriented approaches. There is the absence of a theoretical perspective/framework on KM practices used in each study. Highlights of the summary of the literature analysis are presented in Table 3.

The majority of the studies indicated, have mainly discussed one of the four approaches in KM Practices. Only Rao & McNaughton [56] use technology- and goal-oriented approaches in KM practices. In their study, Knowledge Broker was developed to drive knowledge acquisition, knowledge creation, and knowledge sharing, which helped bridge the gap in the availability of tools and technologies that are needed to support the collaboration and sharing of resources. In terms of goals-oriented KM, the Knowledge Broker promoted both the individual's and the organization's performances by ensuring that the appropriate knowledge is being used in the correct format and manner, and at the right time and place. As a result, the Knowledge Broker has the potential to increase the efficiency of solving regional issues on disaster management.

The table clearly shows that the majority of articles use a technology-oriented approach to KM practices. This is not surprising, considering that information technology has a significant role in accommodating the KM process. Yet, there remains a paucity of research that discuss KM practices in disaster situations with a process-oriented approach, which is what bridges the gap between people and technology-oriented KM practices.

In terms of the disaster management cycle, the majority of articles use KM practices in only one phase, being either pre-, during, or post-disaster. Several studies that include KM practices in the entire cycle of disaster management are those that use technology-oriented KM practices, such as Dufty [68], Kaewkitipong [69], Othman & Beydoun [65], Othman & Beydoun [67], and Othman et al. [66]. In this circumstance, the use of technology-oriented KM practices facilitates the improvement of the sharing of knowledge. Further, such practices improve the combination and matching of various DM activities with which to manage any disaster in the best way possible. The present authors also present a conceptual/theoretical background of the disaster management cycle in this paper.

4.2.1. *People-oriented KM practices in disaster management*

Concerning people-oriented KM practices, the analysis of results reveals that at least 10 of the 72 studies that were analysed reveal that the people-oriented KM has been promoted for some time. Notwithstanding, its application in the disaster context has not been sufficiently evolved. A summary of the articles can be seen in Table 4.

From the people-oriented perspective, this review has identified that KM practices in disaster management supported the improvement of knowledge processes, particularly regarding knowledge sharing. The process of sharing knowledge has a crucial role in ensuring that the experiences gained during a disaster become lessons learned for more effective efforts in dealing with future events.

Several studies have shown that people-oriented KM practices could contribute to increasing individual motivation to build a culture of knowledge sharing that is aimed at reducing disaster risk [20–23]. Research by Ahmad et al. [24] showed that the attitude of sharing knowledge with disaster victims could effectively expand the process of sharing knowledge. As a consequence, sharing knowledge among victims who are part of the knowledge management process will manifest a positive change in people's attitudes towards them helping each another. Through knowledge sharing, activities between victims will strengthen their relationships.

People-oriented KM also affords an increasing awareness and the shaping of individual decisions that are accurate, while preventing inappropriate reactions in responding to disasters. An individual who has positive feelings about their knowledge may allow that to encourage them to share knowledge, in that, individual feelings consist of a salient belief. In a disaster context, several salient beliefs have been affirmed to affect disaster preparedness [25–27].

Results of studies by Bosher et al. [28] and Opdyke et al. [29], demonstrate that KM practices play a significant role in developing knowledge and awareness of the importance of an integrated disaster risk management approach in the post-disaster phase. This knowledge includes potential hazards and increases awareness regarding safety, security, and sustainability issues in order to facilitate more effective disaster risk management strategies.

Table 3
Matrix of KM practices in disaster management.

Authors	Pre-Disaster				During Disaster				Post-Disaster			
	Pe	Pr	T	G	Pe	Pr	T	G	Pe	Pr	T	G
Ahmad et al. (2015)					X							
Ali et al. (2020)										X		
Amaratunga et al. (2018)												X
Becker et al. (2008)	X											
Becker et al. (2013)	X											
Bevilacqua et al. (2014)			X									
Blake et al. (2019)							X				X	
Bosher et al. (2007)									X			
Burastero et al. (2005)			X									
Busapathumrong (2013)										X		
Chua (2005)	X				X							
Chung et al. (2010)			X									
Clarke et al. (2014)			X									
Daneshgar et al. (2011)											X	
Dorasamy et al. (2017)			X									
Dufty (2016)			X				X				X	
Fang et al. (2014)			X									
Foresti et al. (2015)			X									
Fujieda & Kobayashi (2013)										X		
Germain (2010)						X						
Gimenez et al. (2017)							X					
Ha (2018)		X										
Haruyama & Taresawa (2014)	X											
Hashim et al. (2015)							X					
Hirunsalee & Kanegae (2010)	X											
Hristidis et al. (2010)			X									
Ikeda et al. (2016)	X											
Inan et al. (2018a)			X									
Inan et al. (2018b)			X				X					
Jameson & Baud (2016)				X								
Jeong et al. (2016)			X									
Kaewboonma et al. (2014a)			X									
Kaewboonma et al. (2014b)			X									
Kaewkitipong et al. (2016)			X				X				X	
Koria (2009)												X
Kotaki (2015)							X					
Kuppuswamy & Rajarathnam (2013)			X									
Lauer & Matera (2016)						X						
Leskens et al. (2019)			X									
Miles et al. (2019)											X	
Mohideen & Dorasamy (2018)												X
Murphy & Jennex (2006)							X					
Musungu (2016)			X									
Mynett & Vojinovic (2009)			X									
Nakanishi & Black (2018)	X											
Oba et al. (2010)												X
Odoni & Lane (2010)			X									
Oktari et al. (2015)	X											
Opdyke et al. (2017)											X	
Opdyke et al. (2018)									X			
Orchiston & Higham (2016)								X				X
Othman & Beydoun (2013)			X				X				X	
Othman & Beydoun (2016)			X				X				X	
Othman et al. (2014)			X				X				X	
Pathirage et al. (2012)	X											
Peng (2018)				X				X				
Raman et al. (2018)			X				X					
Rao & McNaughton (2019)				X			X					X
Rostis (2007)								X				
Rydstedt et al. (2017)	X											
Sacco et al. (2012)			X				X					
Seneviratne et al. (2010)	X											
Smirnov et al. (2007)							X					
Spiekermann et al. (2019)				X				X				X
Srivastava (2012)	X											
Von et all. (2008)	X											
Wang et al. (2019)	X											
Wang et al. (2020)			X									
Weichselgartner & Pigeon (2015)				X								
Yahya et al. (2016)	X											
Yang & Wu (2019)							X				X	
Zulean & Prelipcean (2013)			X									

Pe: People; Pr: Process; T: Technology; G: Goal.

Table 4
Literature on people-oriented KM practices in disaster management.

Author (s)	Year	Research aim/objectives	Findings	Key Theme	DM Phase
Haruyama & Taresawa [20]	2014	Analysed determinants included among the non-structural measures and differences within social activities related to disaster mitigation in regard to the 2011 mega-tsunami	Knowledge sharing in local community actions is one of the useful critical roles in risk reduction with respect to the tsunami	Willingness of people in building shared knowledge	Pre-Disaster
Rydstedt et al. [21]	2017	Investigated attitudes towards an understanding of the sharing of experience and knowledge of extreme weather damage	KM generates parallel feedback loops as opposed to sharing knowledge that could be used in the form of "lessons learned" to embellish and promote prevention work in the face of potential future damage and losses.	Willingness of people in building shared knowledge	Pre-Disaster
Pathirage et al. [22]	2012	Explored key knowledge factors related to the disaster management cycle and looked at some challenges associated with identified disaster knowledge factors	KM could play a vital role by ensuring the availability and accessibility of accurate and reliable information on disaster risk when needed and by providing useful lessons learned.	Willingness of people in building shared knowledge	Pre-Disaster
Seneviratne et al. [23]	2010	Identified and mapped vital knowledge success factors for managing disasters with success by obtaining the best practices and lessons learned	Identifying critical success factors of KM would be an enabler for successful disaster management	Willingness of people in building shared knowledge	Pre-Disaster
Ahmad et al. [24]	2015	Examined the determining factors of knowledge sharing intentions during a disaster.	The attitude of sharing knowledge with flood victims in Malaysia could effectively expand the process of sharing knowledge	Willingness of people in building shared knowledge	- Pre-Disaster - During Disaster
Becker et al. [25]	2013	Determined the different hazards and preparedness-related beliefs and articulate how public education influences them to increase preparedness	Invented different hazards and preparedness-related beliefs and explained how public education influences them to enhance preparedness	Providing individuals with a correct choice of action	Pre-Disaster
Oktari et al. [26]	2015	Evaluated current efforts of disaster education and developed a conceptual model for enhancing coastal community resilience	The developed model includes the knowledge management aspect to provide individuals with a correct choice of actions that could save lives.	Providing individuals with a correct choice of action	Pre-Disaster
Chua [27]	2005	Compared the two Katrina and Rita disaster preparedness and emergency response efforts from the KM perspectives	The importance of KM to ensure adequate preparation for managing significant natural hazards and to avoid chaotic reactions during a major disaster caused by natural hazards	- Willingness of people in building shared knowledge - Providing individuals with a correct choice of action	- Pre-Disaster - During Disaster
Bosher et al. [28]	2007	Developed a knowledge database and decision support framework to facilitate more effective disaster risk-management strategies from a construction perspective	Knowledge and awareness of integrated approaches is needed for disaster risk-management activities in construction sector	Developing disaster recovery awareness	Post-Disaster
Opdyke et al. [29]	2018	Examined the current state of education and training in construction for post-disaster shelter and housing, and to assess how households gain new construction practice knowledge.	Households gained knowledge via: 1) formal training methods that include reflective observation, active experimentation, and practical experiences, or 2) observation of on-site construction activities.	Developing disaster recovery awareness	Post-Disaster

4.2.2. *Process-oriented KM practices in disaster management*

Content analysis shows that a total of 15 articles (see Table 5) are in the category of process-oriented KM practices, most of which address knowledge acquisition and knowledge conversion. In general, this knowledge conversion process seeks to integrate local wisdom and put the knowledge into practice.

Several studies have shown that KM practices, those that are primarily process-oriented, play a significant role in optimising the use of local and indigenous knowledge. This local and indigenous knowledge uniquely belongs to a particular community/culture that has been developing for a long time as a result of a reciprocal relationship process between the society and its environment. In the context of disaster, local and indigenous knowledge plays a role in increasing preparedness, response, and the capacity to adapt [30–32].

Ikeda et al. [33] and Srivastava [34] emphasise that local and indigenous knowledge is easier to understand by the community; therefore, a thoroughly designed approach is needed to manage this knowledge with the aim of reducing disaster risk. One way of managing this is by combining indigenous and scientific knowledge and methods for transferring/communicating this knowledge.

Another practice of KM process-orientation during a disaster was demonstrated in a study by Lauer & Matera [35], the results of which confirmed that indigenous knowledge plays a role in building individual adaptability to detect ecological changes following a disaster. This point is also supported by Fujieda & Kobayashi [36], whereby indigenous knowledge which is accumulated over long periods of adaptation can increase community resilience in facing disasters.

Other articles cover the process of knowledge conversion, which combines different sets of explicit knowledge to be compiled into knowledge management. Explicit knowledge is the knowledge collected and translated into a form of documentation so that it is easier for others to understand and share. Meanwhile, tacit knowledge is knowledge in a person's brain or mind according to the person's understanding and experience. Usually, this knowledge is unstructured, difficult to define, and conveys informal language to others, and its content includes personal understanding. Studies conducted by Nakanishi & Black [37] and Wang et al. [38] emphasise that both explicit and tacit knowledge is required for developing more effective disaster mitigation and preparedness. Notwithstanding, process-oriented KM practices assist in capturing, managing, and disseminating significant amounts of data and knowledge during a disaster [39].

Other studies show that KM practices also encourage the formation of actionable knowledge. In the context of a disaster, this actionable knowledge is required and critical to initiate an immediate and appropriate response [40,41]. Meanwhile, in the post-disaster phase, process-oriented KM practices contribute to the smooth running of the transition process between the recovery and post-disaster phases and build up the social recovery resilience [42,43].

4.2.3. *Technology-oriented KM practices in disaster management*

The use of technology-oriented KM in disaster management has provided a significant contribution in the disaster management field. This topic remains of ongoing interest among scholars, as indicated by 38 articles from a total of 72 studies, as presented in Table 6.

The results of the analysis show that technology-oriented KM is implemented by building numerous information systems that assist in

Table 5
Literature on process-oriented KM practices in disaster management.

Author (s)	Year	Research aim/objectives	Findings	Key Theme	DM Phase
Ha [30]	2018	Studied how to apply indigenous knowledge to natural hazard preparedness as a component of disaster management	The discipline of disaster preparedness must adjust its indigenous knowledge towards sustainable measures	Incorporate local and indigenous knowledge	Pre-Disaster
Becker et al. [31]	2008	Examined a specific case study in which a traditional Native American tale was combined with modern hazard mitigation techniques to create a tsunami hazard educational video	Local and indigenous knowledge could be employed effectively to initiate hazard education and improve response to warnings	Incorporate local and indigenous knowledge	Pre-Disaster
Hirunsalee & Kanegae [32]	2010	Explored the power of local knowledge, which is useful to promote community self-protection and to cope with disaster in the flood-prone residential area on Ayutthaya Island	Local knowledge is not only transferred by the family or via kinship (i.e., strongest-tie social network) but also by the tutor weaker-tie (i.e., social network) with the more appropriate effectiveness regarding some of the knowledge	Incorporate local and indigenous knowledge	Pre-Disaster
Ikeda et al. [33]	2016	Explored a knowledge-based approach directed on non-structural measures to decrease glacier lake outburst floods (GLOF) risk	The importance of communicating disaster risks in a clear way, while supporting the attempts of residents to introduce novel scientific knowledge into existent local knowledge	Incorporate local and indigenous knowledge	Pre-Disaster
Srivastava [34]	2012	Highlighted concerns related to managing indigenous and scientific knowledge and the method of transferring indigenous knowledge from one community to the next to reduce risk in communities living alongside multiple hazards/ disasters in regions of India that are prone to cyclones, floods, earthquakes and drought	Indigenous knowledge is more understood/implied; therefore, a thoroughly designed approach would be needed to manage this knowledge	Incorporate local and indigenous knowledge	Pre-Disaster
Lauer & Matera [35]	2016	Evaluated individual differences in the Solomon Islanders' ability to detect ecological change following a major tsunami in local, shallow-water marine environments	The role of indigenous knowledge in building adaptation at the local level to detect ecological change following a disaster	Incorporate local and indigenous knowledge	During Disaster
Fujieda & Kobayashi [36]	2013	Explored the potential use of traditional Fijian housing as an alternative living environment restoration methodology	From the resilience perspective, indigenous knowledge which is generated/accumulated over long periods of time of adaptation to the local environment can potentially augment the capacity of the local community to overcome disasters	Incorporate local and indigenous knowledge	Post-Disaster
Nakanishi & Black [37]	2018	Explored the role of tacit and explicit knowledge in disaster planning and evacuation for flood events with a case study of Takamatsu, Japan	Both explicit and tacit knowledge are required for local-area evacuation plans and to ensure that Japan's land-use planning practitioners can play a crucial role in disaster mitigation	Knowledge conversion of disaster data into DRR knowledge	Pre-Disaster
Wang et al. [38]	2019	Assessed the knowledge service platform and its effectiveness in disseminating earthquake-related disaster risk reduction (DRR) knowledge to decision makers and to the general public	Over the long term, the methodology would 1) support the conversion of disaster data into disaster risk reduction (DRR) knowledge and 2) provide services for international organizations, government institutions, research institutions, educational institutions, businesses, and also at the level of the general public	Knowledge conversion of disaster data into DRR knowledge	Pre-Disaster
Germain [39]	2010	Examined knowledge sharing in crisis situations	Knowledge sharing activities through the role of HRD in emergencies, such as the implementation of inter-agency collaborative planning and the establishment of comprehensive processes for collecting, managing, and disseminating primary themes, facts that are gleaned from practical scenarios, and also prior practical knowledge	Knowledge conversion	During Disaster
Von et al. [40]	2008	Illustrated the incorporation of updated knowledge and KM into the framework of network-centric operations and network-enabled capabilities by employing Boyd's Observe, Orient, Decide, and Act (OODA) loop-based decision-making in complex, settings that are changing in a dynamic way	Knowledge must be collected, combined with information produced during the disaster, and converted into actionable knowledge	Put knowledge into practice	Pre-Disaster
Yahya et al. [41]	2016	Defined processes that are typical and central to transferring knowledge from multiple transmitters to a unique receiver, and their architecture was star topology	The utilization of the essential processes of knowledge transfer would equip stakeholders with a tangible means of transferring knowledge by systematizing the knowledge transfer process itself	Put knowledge into practice	Pre-Disaster
Ali et al. [42]	2020	Employing a 2010 flood case study in Sindh, Pakistan, distinguished knowledge management as one of the factors that facilitated the transition process between relief, rehabilitation, and recovery	Mechanism to maintain, distribute and gather data during the Post-Disaster Recovery (PDR) process to assist in future planning because of an uncoordinated and fragmented transition between long-term recovery phases	Smooth transition for disaster recovery	Post-Disaster
Busapathumrong [43]	2013	Explored Thailand-based disaster management regarding the impact of a disaster on the subpopulations and the vulnerability/resilience of females, infants, old-aged citizens, and disabled individuals	The utilization of the KM model in Thailand-based disaster management and the building up of the social recovery resilience	Knowledge-based resilience	Post-Disaster

Table 6
Literature on technology-oriented KM practices in disaster management.

Author (s)	Year	Research aim/objectives	Findings	Key Theme	DM Phase
Dorasamy et al. [44]	2017	Conducted a systematic review of articles published over the past two decades on the applicability of a knowledge-driven system for crisis management and designed the integrated Community Emergency Management and Awareness Systems (iCEMAS)	- In support of the crisis management effort, the knowledge management system supports the creation of dynamic systems - Combining Knowledge Management System (KMS) into Emergency Management Information Systems (EMIS) will enable people to create adjustments to their roles based on the situation	Role of ICT in building awareness and preparedness	Pre-Disaster
Inan et al. [45]	2018a	Introduced an Agent-Based (AB) methodology for the analysis of knowledge in converting Disaster Management Plans (DISPLANs) into a compilation of knowledge units that can be stored within a consolidated repository	The use of knowledge management enables the combinations of knowledge between differing plans to gain the appropriate strategy promptly and sharing actions between plans	- Role of ICT in building awareness and preparedness	Pre-Disaster
Musungu [46]	2016	Assessed knowledge production processes by applying a nodal governance framework to study the creation of survey data/geographic information system (GIS) knowledge among different stakeholders in two different flood-prone informal settlements in Cape Town, RSA and involved the academic researchers, the local government, the local communities, and also the non-profit organizations	The importance of local knowledge production for effective governance and needs to consider the involvement of various stakeholders	- Role of ICT in building awareness and preparedness	- Pre-Disaster
Bevilacqua et al. [47]	2014	Studied information flow and its distribution in risk management, specifically hydrological hazards by the IDEF0 modelling tools	The modelling tools improve risk management aspects that are more tangible such as information flow, because of the modelling language that is easy to use makes it possible to be used by all actors involved in emergency management	- Role of ICT in building awareness and preparedness	- Pre-Disaster
Foresti et al. [48]	2015	Developed ASyEM, which is an advanced system for emergency management and it combines the potential in mobile social data and communication with smart sensors	The application of KM would increase the reliability and the efficiency of the entire situational awareness services, limit the critical areas and thus obtain pertinent information regarding disaster emergency response and the accomplishment of search and rescue operations	- Role of ICT in building awareness and preparedness	- Pre-Disaster - During Disaster
Odoni & Lane [49]	2010	Advocated a knowledge-based approach-theoretic hydrological modelling rather than data-theoretic for flood risk management	Hydrological modelling fits into the modern realisation that the process by which the public engages in knowledge needs to be pushed upstream	- Role of ICT in building awareness and preparedness	- Pre-Disaster
Hristidis et al. [50]	2010	Surveyed and synthesized the cutting-edge knowledge regarding data management and data analysis in disaster scenarios, as well as identify the challenges and directions for future research	The crucial role of knowledge management has been in devising ways of managing and analysing the data produced in disaster situations	- Role of ICT in building awareness and preparedness	- Pre-Disaster
Kuppuswamy & Rajarathnam [51]	2013	Highlighted the factors that influence the attitude and purpose of the coastal community to use information and communication technologies (ICT) for disaster and growth communication in Tamilnadu, within the Nagapattinam district of India	ICT has great potential in disaster management as technological support is required in all stages to collect data, increase preparedness, communicate, disseminate warnings, knowledge management, and build the capacity	- Role of ICT in building awareness and preparedness	- Pre-Disaster
Raman et al. [52]	2018	Examined the role of knowledge management systems (KMS) for disaster planning and response regarding social work in Malaysia	The role of Information Technology (IT) in disaster management and the willingness to share knowledge during a disaster situation	- Role of ICT in building awareness and preparedness - Role ICT in emergency response	- Pre-Disaster - During Disaster
Inan et al. [53]	2018b	Supported the use of the Decision Support System (DSS) that could be used as a single access point for disaster management	Information elicitation and retrieval processes could be used in the decision-making process for stakeholders during an emergency response	- Role of ICT to support decision-makers - Role ICT in emergency response	- Pre-Disaster - During Disaster
Murphy & Jennex [54]	2006	Discussed the use of knowledge management for emergency information systems	Emergency information systems are improved by the integration of knowledge management tools and principles	Role ICT in emergency response	- During Disaster
Kotaki [55]	2015	Described numerous lessons for initial and emergency response from the Great East Japan Earthquake	The importance of establishing a system that would allow the government or the local government to adopt temporary disaster emergency measures, such as early recovery, evacuation, rescue, emergency medical rescue, and stocking of emergency supplies	- Role ICT in emergency response	- During Disaster
Rao & McNaughton [56]	2019	Identified the work that been undertaken to assist in bridging this gap via the design and development of Knowledge Broker, which is a technological solution for the recognition and exchange of Knowledge Resources that could be shared through different locations (e.g., countries)	The Knowledge Broker could likely increase the productivity of regional issues through the exchange of knowledge resources	- Role ICT in emergency response	- During Disaster
Smirnov et al. [57]	2007	Developed an approach to in disaster-response operational decision-making using ontology-driven knowledge sharing and the application of well-research tasking exercises from the field of production network management that promote the	The importance of the KM concept in production networks to the entirely disaster relief and humanitarian logistics new areas	- Role of ICT in decision-making	- During Disaster

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Table 6 (*continued*)

Author (s)	Year	Research aim/objectives	Findings	Key Theme	DM Phase
		use of existing methods and tools for problem solving			
Blake et al. [58]	2019	Analysed how the transport system, networks, and supply chain reacted and responded to earthquake-related disturbances	Distribution and use of knowledge in post-disaster decision-making that is related to response and recovery across all modes of transport, with strengthened information sharing, data sharing, and decision-making	- Role of ICT in decision-making	- During Disaster - Post-Disaster
Daneshgar et al. [59]	2011	Introduced the conceptual model of knowledge – sharing in post-disaster situations, namely Awareness Information Network (AIN)	Knowledge sharing is essential for the management of post-disaster situations in which natural events are interrupted by mostly unpredictable, disruptive, and life-threatening disasters	- Role of ICT in decision-making	- Post-Disaster
Miles et al. [60]	2019	Studied the facilitation of a community of practice (CoP) for disaster recovery modelling	CoP assists in the generation and appropriation of a shared symphony of novel ideas, novel approaches, and institutional records in order to simulate lifeline infrastructure in the form of comprehensive and meaningful socio-technical systems and in ways that are pertinent during any processes of recovery	- Role of ICT in decision-making	- Post-Disaster
Opdyke et al. [61]	2017	Examined the inter-organizational networks that are moulded to coordinate resources for post-disaster infrastructure recovery	Inter-organizational resource coordination assists organizations in strengthening the efforts made by humanitarian response systems via accommodating the resource allocation and knowledge sharing between the different organizations	- Role of ICT in decision-making	- Post-Disaster
Kaewboonma et al. [62]	2014a	Investigated the ontology modelling approach for the management of drought	The importance of knowledge gathered, categorized, and incorporated into a decision-making system for the management of drought	- Role of ICT to support decision-makers	- Pre-Disaster
Kaewboonma et al. [63]	2014b	Examined the incorporation of data on drought management for the creation of ontology-based semantic analysis to support decision-makers at different stages of a disaster	The visualization of knowledge offers robust accessibility and decision support systems for drought management information	- Role of ICT to support decision-makers	- Pre-Disaster
Leskens et al. [64]	2019	Examined how the use of interactive flood simulation models influence decision-making	Concentrating on the collective information construction process is a useful perspective for recognizing how concept results are combined with practitioners' expertise towards a common understanding of the situation at hand	- Role of ICT to support decision-makers	- Pre-Disaster
Othman & Beydoun [65]	2013	Developed a Disaster Management Metamodel that can assist as a representative layer of DM expertise	The Disaster Management Metamodel promotes the sharing of knowledge in a better way and promotes the combination and matching of various DM activities to manage any disaster in the best way possible	- Metamodel-based disaster management	- Pre-Disaster - During Disaster - Post-Disaster
Othman et al. [66]	2014	Developed and validated a Disaster Management Metamodel (DMM) that enables knowledge sharing and successful activities in the pre-disaster process	DMM allows knowledge sharing and effective activities in pre-disaster phase	- Metamodel-based disaster management	- Pre-Disaster - During Disaster - Post-Disaster
Othman & Beydoun [67]	2016	Presented a metamodel-based architecture suitable for a variety of shared knowledge settings	A knowledge-based system approach is used to address the challenge of knowledge sharing to facilitate the structuring, storage, and re-use of knowledge on disaster management	- Metamodel-based disaster management	- Pre-Disaster - During Disaster
Dufty [68]	2016	Examined the current body of literature to identify the primary utilization of Twitter in Australia and overseas in crisis management over the past ten years	- The usefulness of Twitter as a tool to identify hazard risk, engage the community in disaster mitigation and preparedness, communicate an early warning, and as a tool to allocate resources during disaster recovery - The essential uses of Twitter as a crowdsourcing platform to provide real-time information, emotional support, and recognize the needs and vulnerabilities of affected communities	- The use of social media	- Pre-Disaster - During Disaster - Post-Disaster
Kaewkitipong et al. [69]	2016	Used a qualitative case study to analyse the information sharing activities using social media within local communities to combat the 7-month Thai flood crisis of 2011	The potential utility of social media as a knowledge sharing tool in disaster management	- The use of social media	- Pre-Disaster - During Disaster - Post Disaster
Hashim et al. [70]	2015	Described the determinants that affect the goodness of fit between social media and information sharing activities during a disaster	Social media was one of the most commonly used ICT resources for sharing information in the event of a disaster	The use of social media and virtual communities of practices	- During Disaster
Gimenez et al. [71]	2017	Presented a Delphi study that was undertaken by a multidisciplinary group of international practitioners working in disaster management organizations to assess the role of virtual communities of practice (VCoPs) that drives organizations to advance the disaster management	Virtual communities of practice (VCoPs) leads organizations to improve the management of disasters	The use of social media and virtual communities of practices	- During Disaster

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Table 6 (*continued*)

Author (s)	Year	Research aim/objectives	Findings	Key Theme	DM Phase
Burastero et al. [72]	2005	Developed STORM3 as an innovative management and monitoring system with a real-time forecast and warning of flood risk	The use of Decision Support System tools that facilitate knowledge sharing within the public protection centres will reduce the risk level in a region	- Forecasting for disaster mitigation	- Pre-Disaster
Chung et al. [73]	2010	Studied the importance to have linking information and data generated by the system for the process of making a decision in mitigating the impact of flood	Flood mitigation using a human-centred system with a cyber-infrastructure framework	- Forecasting for disaster mitigation	- Pre-Disaster
Clarke et al. [74]	2014	Reviewed the existing flood risk system and methods	The use of KM assisted the conversion of geographic information systems from complex server-dependent into a more flexible and user-friendly web-browser-based tool to be utilized in managing flood incidents	- Assist in converting complex to user-friendly system	- Pre-Disaster
Jeong et al. [75]	2016	Researched the implementation of the Smart Big Board (SBB) program using the technique of service design, which is commonly used to improve public services	The use of KM for user-friendly disaster management in response to different and complex forms of disaster	- Assist in converting complex to user-friendly system	- Pre-Disaster
Fang et al. [76]	2014	Performed an integrated study regarding managing water resources with a basis employing geoinformatics and using Remote Sensing, GIS, GPS, Enterprise Information Systems, and also cloud services	A prototype IIS (i.e., Water Resource Management Enterprise Information System, WRMEIS) can forecast flood risk using the data acquisition, data management, data sharing, data modelling, and knowledge management	- Assist in converting complex to user-friendly system	- Pre-Disaster
Wang et al., [91]	2020	Built systems for sharing information and providing thematic DRR services, based on the formulation of disaster meta-data standards using the Wenchuan earthquake that occurred ten years ago	This system prevailed in changing the paradigm from disaster data to the provision of knowledge services and actualizing the integration and sharing of knowledge resources (i.e., data, maps, expert opinions, research literature, and educational material)	- Assist in converting complex to user-friendly system	- Pre-Disaster
Mynett & Vojinovic [77]	2009	Studied the role of urban hydro-informatics in assessing the impact of climate change on urban floods and health risks in the context of the UN Millennium Development Goals and represented in a case study for Dhaka, Bangladesh	Hydro-informatics is also used in the fields of knowledge systems and knowledge management	- Speed up access to information and knowledge	- Pre-Disaster
Sacco et al. [78]	2012	Studied the use of dynamic taxonomy databases during geo-hydrological emergencies and compared the technique to the well-used relational database query methods	Dynamic taxonomies are a general KM model for complex configuration for rapid and accurate searches, one of the essential factors for timely response to emergencies	- Speed up access to information and knowledge	- Pre-Disaster - During Disaster
Zulean & Prelipcean [79]	2013	Examined the risks and organizations coping with disasters, assessed emergency preparedness, used Knowledge Management Systems and Foresight Methodology methods, and identified shortcomings of the Romanian Emergency System	Integrated management should be the solution for addressing the institutional gap and improving communication	- Speed up access to information and knowledge	- Pre-Disaster
Yang & Wu [80]	2019	Developed an earthquake response and recovery taxonomy using online information resources to organize and share earthquake-related information that is found online	The taxonomy is relevant to online information earthquake management, plus knowledge and disaster management	- Speed up access to information and knowledge	- During Disaster - Post-Disaster

creating a disaster plan by involving various stakeholders, so that it is possible to achieve the right strategy and the sharing of actions between plans [44–46]. The utilization of ICT in disaster management is also widely used as a modelling tool to minimize disaster risk and to facilitate an appropriate emergency response process [47,48,50].

ICT has a great potential in disaster management because technological support is required at all stages to collect data, increase preparedness, communicate, disseminate warnings, knowledge management, and to build the capacity. To this end, technology-oriented KM also play a role in devising ways of managing and analysing the data that are produced in disaster situations [51–56]. Consequently, the various facilities that are offered from the use of technology-oriented KM can strengthen information sharing, data sharing, and effective decision-making [57–64].

Another topic that has become the centre of attention among researchers is the Disaster Management Metamodel development. The development of this methodology promotes the sharing of knowledge in a better way. It improves the combination and matching of various DM activities to manage any disaster in the best way possible [65–67].

KM practices in disaster situations also encourage more effective use of social media. Some scholars have confirmed the usefulness of social media as a tool to identify hazard risk, knowledge sharing, engagement of the community in disaster mitigation and preparedness, communication of an early warning, and as a tool to allocate resources during disaster recovery [68–71].

Other studies show the use of KM in gaining an innovative management and monitoring system with a real-time forecast and warning of disaster risk [72,73], and facilitating the conversion of a complex server-dependence into a more flexible and user-friendly information system to be utilized in disaster management [74–76]. Studies also confirm that technology-oriented KM allows the speedy access to information/knowledge, therefore, this will contribute to the solution for addressing the institutional gap and improving communication [77–80].

4.2.4. *Goals-oriented KM practices in disaster management*

In contrast to technology-orientation, studies that discuss goals-oriented KM practices still receive little attention from researchers. Only 11 out of 72 articles have addressed the topic, as shown in Table 7. This condition is not surprising, considering that the orientation of KM is solely focused on people, processes, and technology. The goals-oriented approach to KM is still regarded as newly developed.

Goal-oriented KM practice in disaster situations has an orientation towards improving the performance of both individuals and institutions engaged in disaster, with the primary objective of creating a more disaster-resilient community. The KM approach is applied to ensure that appropriate knowledge can be used in the right format, mode, time, and place, therefore, this would help to resolve the problems of disaster risk

Table 7
Literature on goals-oriented KM practices in disaster management.

Author (s)	Year	Research aim/objectives	Findings	Key Theme	DM Phase
Rao & McNaughton [56]	2019	Identified the work that been undertaken to assist in bridging this gap via the design and development of Knowledge Broker, which is a technological solution for the recognition and exchange of Knowledge Resources that could be shared through different locations (e.g., countries)	The Knowledge Broker has the potential to increase the efficiency of solving regional issues through the sharing of knowledge resources	Contribute to address problem	- Pre-Disaster - Post-Disaster
Weichselgartner & Pigeon [81]	2015	Examined insights into emerging knowledge conceptualizations that would advance science and policy in disaster	Improving the incorporation of various scales, different societal players, diverse knowledge sources, and disciplines into disaster risk research would increase its importance for decision-making in policy and practice	Contribute to address problem	Pre-Disaster
Jameson & Baud [82]	2016	Investigated how variations of knowledge about flood risk lead to a more effective governance structure of flood management (FM) in Chennai, India	Transforming knowledge shows a strong potential for linking long-term water management strategies which in turn would help to resolve the problems of flood risk and drinking water scarcity in the city	Contribute to address problem	Pre-Disaster
Spiekermann et al. [83]	2019	-Analysed knowledge fragmentation and its causes at different phases in the disaster management using two case studies in Austria and Mexico	Understanding different facets of knowledge strengthens individual and institutional learning, and determining the necessary social and functional adjustments with which to address crowding issues of disaster risk reduction	Contribute to address problem	- Pre-Disaster - During Disaster - Post-Disaster
Peng [84]	2018	Assessed flood-prone areas, a correlation analysis of flood factors for quantitative risk assessment was identified to assist in the further management of disaster prevention	Using a KM approach could support local governments to understand the risk level of the different administrative areas to help them develop effective mitigation strategies	Contribute to address problem	- Pre-Disaster - During Disaster - Post-Disaster
Rostis [85]	2007	Explored the validity of the learning practical experience approach to knowledge management and suggested a number of alternative methods for improving potential outcomes	The knowledge management approach motivates the organization to change or adapt to ensure the mistakes or oversights are highlighted as lessons and are not repeated	Contribute to address problem	- During Disaster
Amaratunga et al. [86]	2018	Defined gaps in the knowledge base of construction professionals that hinder their ability to help create a more disaster-resilient community	Knowledge management as one of the key knowledge gaps in the main body of disaster resilience concepts in the construction process	Contribute to address problem	- Post Disaster
Mohideen & Dorasamy [87]	2018	Conducted action research to improve the disaster recovery process in a multinational firm in Malaysia	Such an economic structure could benefit from KM concept inclusion within the improvement process to underpin its activities in disaster recovery	Contribute to address problem	- Post Disaster
Oba et al. [88]	2010	Explored the needs of potential enhancement regarding an adaptation process and examined the effectiveness of such a program for victims of flooding and landslide in the Lublae district of Uttaradit Province, Thailand with respect to their adaptation to all stages of a disaster and their psychological well-being	The importance of KM on the potential enhancement program (PEP) for the same type of victims in scenarios of flooding and landslides	Contribute to address problem	- Post Disaster
Orchiston & Higham [89]	2016	Explored the outcomes and responses of the tourism sector, outlining conclusions that afford knowledge on the theory and practice of responses to tourism disasters, knowledge management, and recovery marketing	The important consequences of knowledge management and the collaboration and communication between agencies	Effective inter-agency collaboration and communication	Post-Disaster
Koria [90]	2009	Researched the 2004 Indian Ocean tsunami in Sri Lanka and the post-disaster recovery process from the angle of the project and program management (PPM) of a major international non-governmental organization that was involved in the recovery operation	KM is essential to manage innovation in recovery programs that are large in size and complex in nature	Create innovation in disaster recovery	Post-Disaster

[56,81–88].

Implementing KM in a disaster situation confirms the important consequences of knowledge management and effective collaboration and communication between agencies [89]. Disaster management involves programmes that are large in size and complex in nature, hence, in this state KM practices are required to create innovation [90].

5. Future imperatives for KM in disaster

This review had recognized some of the KM practices in disaster management, they are: i) knowledge sharing and providing lessons learned to reduce potential damage/losses; ii) developed a knowledge database/framework for decision support; iii) knowledge capture through reflective observation, active experimentation, and practical experiences; iv) the use of local and indigenous knowledge of disaster; v) communicating scientific disaster knowledge into existent local knowledge; vi) knowledge conversion of disaster data into DRR knowledge; vii) the use of Information Technology (IT), Knowledge Management System (KMS), ontology modelling, the Decision Support System (DSS), knowledge services platform, social media, and metamodel-based disaster management; viii) knowledge sharing through a community of practice (CoP) for disaster.

The aforementioned KM practices had shown several implications in disaster management including i) increase people's ability to create common information, providing individuals with the right option of action; ii) strengthen the role of local and indigenous knowledge; iii) fostering the conversion of disaster data into DRR knowledge and advancing actionable knowledge; iv) promote effective decision-making

Table 8
Outline of further empirical investigation on KM practices in Disaster.

Domain	Latest state regarding research	Potential research directions
Type of natural hazards	Most research is drawn from the experience of a flood disaster	- Studies of KM practices in other hazards and pandemic conditions - Studies of KM practices which consider multiple hazardous events/cascading disasters
Disaster Management Cycle	Majority of articles use KM practices in only one phase, either pre-, during, or post-disaster	- Studies which applied KM within the entire cycle of disaster management - Studies of KM practices which contribute to reducing vulnerability and enhancing capacity in dealing with a disaster
KM practices	Most studies mainly discussed one of the four approaches in KM practices	- Studies which comprehensively applied KM using a people-, process-, technology- and goal-oriented
People-oriented KM practices	- Suggest a subject with less attention - Studies focus on how an individual's knowledge process by gaining experiences during a disaster to be used as future lessons learned	- Studies on the knowledge internalisation to transform "know-what" (cognition) into "know-how" (capacity to act) - Studies to identify enabling component of people-oriented KM in a disaster
Process-oriented KM practices	- Denote a topic of ongoing research attention - Studies concentrate on the conversion of data into DRR knowledge	- Studies on how to translate individual knowledge into collective knowledge - Studies to identify enabling component of process-oriented KM in a disaster
Technology-oriented KM practices	- Indicate a topic of most significant research attention of KM practices - Studies converge on using technology to create disaster plans, collecting and analysing data during disaster, making modelling and forecasting, and using social media and metadata in disaster situations	- Studies to anticipate technological failure in disaster situations - Studies on how people perceive the use of technology during a disaster - Studies to identify enabling component of technology-oriented KM in a disaster
Goal-oriented KM practices	- Appear to suggest a theme with less interest - Studies concentrate on the role of KM in solving problems, streamlining collaboration, and generating innovation	- Studies on transgenerational disaster knowledge transmission (sustaining knowledge) and new knowledge creation - Studies to identify enabling component of goal-oriented KM in a disaster

by transforming complex systems into user-friendly and accelerating access to information and knowledge; and v) contributes to overcoming problems, nourishing coordination, and generate creativity.

Although this review shows various KM practices in managing disasters and their implications, the presented studies have not incorporated theoretical perspectives/frameworks on both Knowledge Management and Disaster Management. Further research is, therefore, needed to provide more reliable power over what Knowledge Management can provide for more effective disaster management; thus, it would bridge the gap between theory and practice.

Based on the content analysis of the body of literature, further empirical investigations of KM in the disaster context have been identified, as presented in Table 8.

Previous studies did not provide a comprehensive architecture for the organization of KM that could likely support the methodology/techniques of KM practices in the disaster management field. Notwithstanding, this structure is needed as a foundation for the various stages of any knowledge management process (these include: creation, acquisition, storage, transfer, distribution, and application) and disaster management stages (these include mitigation, preparedness, emergency response, rehabilitation, and reconstruction).

With these assumptions, the content analysis of "KM practices in disaster management" highlighted the call for a systemic method to the analysis of a taxonomy of a variety of applications and tools, with the aim of ameliorating the efficiency and effectiveness of KM adoption in disaster field.

This gap generates a highly interesting topic within the management literature: the study of the degree of consistency between management strategies and the technologies that are employed. In the context of disaster management, the question of consistency between the nature of organizational knowledge and knowledge management systems (KMSs) that are used to advance knowledge management is hugely important. The consistency between the type of knowledge and the KMSs that are employed is a crucial factor for success that could be the driving force of the KM process. Unfortunately, if there is no alignment among the type of knowledge and KMS, then it will cause inefficiencies.

The research agenda that is discussed in the above, provides recommendations for future study and to describe on the consequences for both academics and practitioners. With respect to the academics, the research gaps found and the following research questions suggested reflect several potential fields of inquiry to strengthen the knowledge base in the area of information management in the event of a disaster. This literature review shows that knowledge management promotes the overall sustainable disaster management effort by advancing its performance at the same time and this concept should be understood and employed by practitioners.

6. Conclusion

This study has analysed a systematic review of the KM practices in disaster management, which identified several gaps in the literature to determine the direction of further research that will advance the body of literature. Concerning the research question, this review intends to assess the use of KM in disaster management and the impacts of KM practices on disaster management performance.

The general statistical information of the literature advises on an overview of the publications incorporated in this study. It has permitted the provision of a summarized perspective of the articles on the topic of KM in the disaster field. The analysis also showed that the pattern of the number documents on the subject has been rising in recent years. The majority of publications are based on the flood disaster experience.

The reviews conducted reveal that the majority of the examples in the body of literature only provide fragmented insights of KM practices in managing the disaster. There is no single article that comprehensively executes KM practices using people, process-, technology- and goal-oriented approaches. The technological-oriented KM practices seem to be dominating research in the disaster management context. Notwithstanding, the body of knowledge regarding the topics of people-, process-, and goal-oriented is somewhat limited.

Hence, the existing empirical studies provide only fragmented insights about KM in disaster management. Given the importance of knowledge to reduce the impact of disasters, the field of knowledge management in disaster management is essential. This research clearly demonstrates that the topic still needs more research, which in turn advises scholars on a variety of research directions.

The analysis of the content of the articles that are included in this study has provided a detailed summary of the main issues that have been incorporated by studies on the application of KM in the disaster management field. It enables to identify the impact of KM practices and any important research gaps regarding the literature on the subject.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The first author gratefully acknowledges the opportunity and support from Integrated Research on Disaster Risk (IRDR), Beijing, China, which has awarded her as one of IRDR Young Scientist. This publication would not have been feasible without the financial support from the Indonesian Ministry of Education and Culture (KEMDIKBUD) under 2020 Doctoral Research Grant No. 174/SP2H/ADM/LT/DPRM/2020.

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